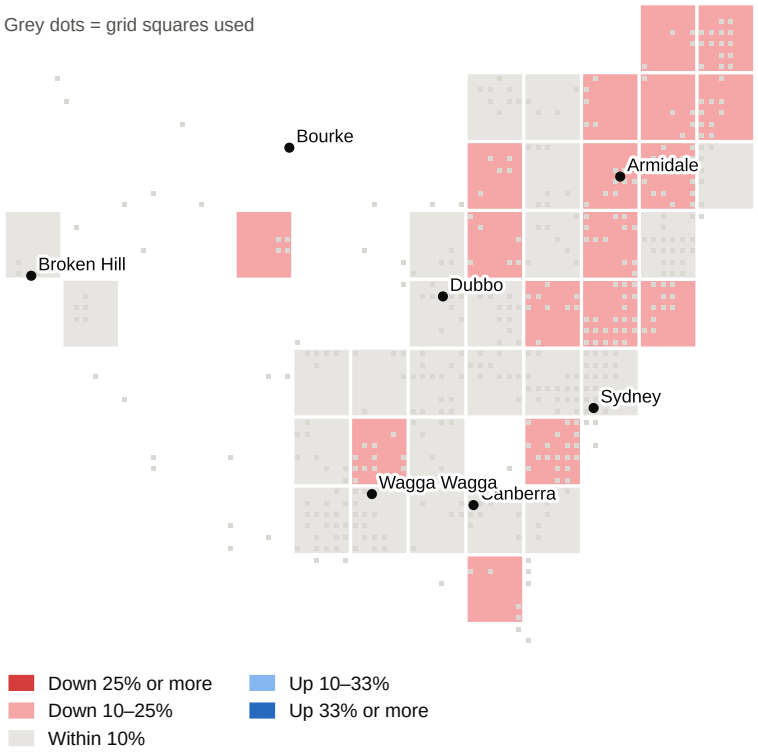
QUESTION 2 · PART A

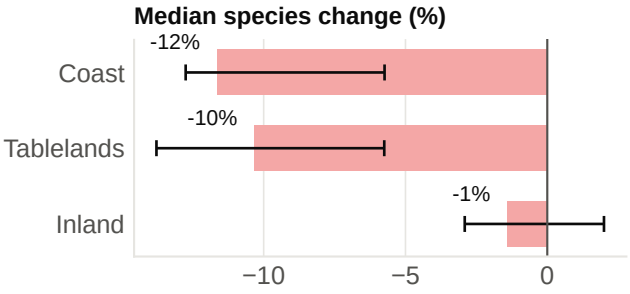
Are declines concentrated in particular regions, habitats or ecological groups?

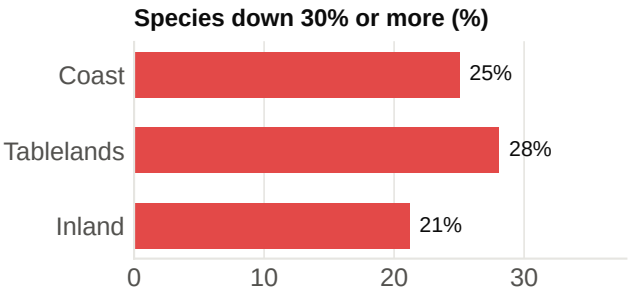
Regions

Declines are widespread, a little steeper on the coast than inland

Median species change in share of squares where recorded, 1977–1990 → 2006–2020 (fair comparison)







KEY FINDINGS

- Typical species change: coast -12%, tablelands -10%, inland -1%.
- 41% of map cells are down 10% or more, scattered across the state rather than clustered.
- Continuing decliners fall hardest: coast 11, inland 9, tablelands 7 species.

POSSIBLE CAUSES · OUR SPECULATION, NOT TESTED HERE

- Coast: housing and infrastructure growth along the coastal strip has replaced woodland and scrub, and the habitat left behind suits Noisy Miners, Common Mynas and other adaptable birds.
- Tablelands: long-cleared grazing country with small, ageing woodland remnants; eucalypt dieback in the 1980s–90s (notably in New England) removed further habitat.
- Inland: much of the clearing happened before 1977, so the starting point was already low. Many inland birds are nomadic and follow rain, so their presence swings rather than falls steadily.
- Prediction: the first report found inland breeding fading. Presence often falls after breeding does, so inland declines may show up in the next decade of surveys.

Atlas surveys 1977–2020; same effort per grid square in each period (432 squares). Change is 1977–1990 to 2006–2020. Region names are not yet confirmed, so zones are used.

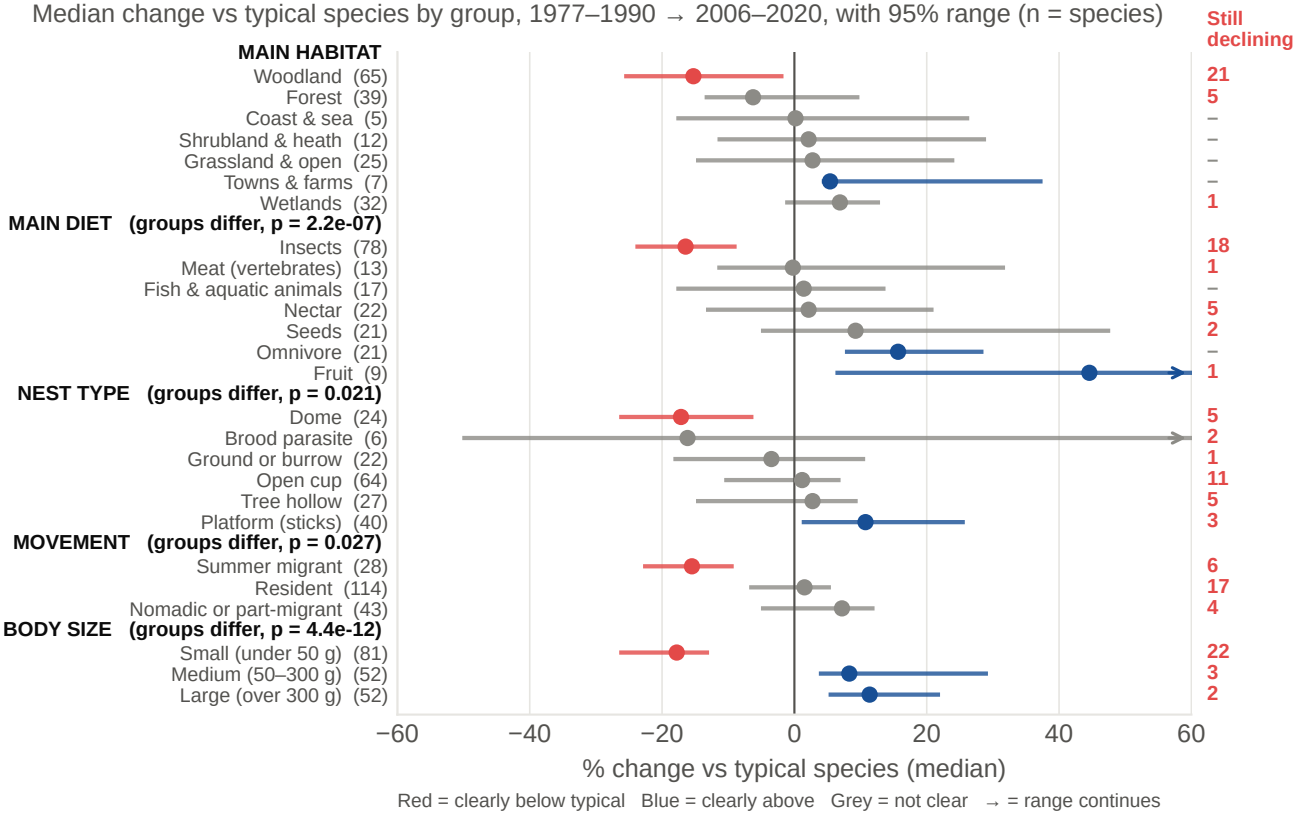
QUESTION 2 · PART B

Are declines concentrated in particular regions, habitats or ecological groups?

Habitats and ecological groups

Declines are concentrated in small, insect-eating woodland birds

Median change vs typical species by group, 1977–1990 → 2006–2020, with 95% range (n = species)



KEY FINDINGS

- Clearly below typical: small birds (-18%), dome nesters (-17%), insect-eaters (-16%), summer migrants (-15%), woodland birds (-15%).
- Clearly above typical: fruit-eaters (+45%), omnivores (+16%), large birds (+11%).
- Groups differ by more than chance for body size, main diet, nest type, movement; habitat differences are suggestive only.

POSSIBLE CAUSES · OUR SPECULATION, NOT TESTED HERE

- Small insect-eaters depend on a shrubby, littered ground layer that grazing, clearing and 'tidying' remove. Their small size also makes them the easiest for Noisy Miners to chase off.
- Dome-nesters build low in shrubs, where nests are exposed to cats, foxes, rats and nest-robbing birds such as currawongs, which do well around people.
- Fruit-eaters, omnivores and large birds gain from gardens, street trees, fruit crops, spilt grain and farm dams: food that people supply year-round.
- Summer migrants face pressure twice: in NSW and on their routes and wintering grounds.
- Prediction: without action, birdlife keeps shifting towards fewer, more adaptable species.

Atlas surveys 1977–2020; same effort per grid square in each period (432 squares). Change is 1977–1990 to 2006–2020. Traits are provisional (species_traits_provisional.csv) and should be checked against the team's trait table.